Offline Writer Identification: possible computer vision based approaches to writer recognition problems

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Abstract :

This report addresses the problem of handwritten text recognition using computer vision methods. The main objective is to identify the author of a text by comparing it with known signatures, employing two distinct approaches based on the visual representation of available data. The first method relies on matching visual features using the Scale-Invariant Feature Transform (SIFT) algorithm, while the second utilizes the Bag of Words (BoW) representation for comparing visual patterns. The report evaluates the effectiveness of these approaches in contexts with limited dataset sizes, also analyzing practical implications and potential avenues for future improvement. The results highlight the challenges and opportunities in the field of handwritten text authorship recognition, thereby contributing significantly to advancements in forensic and historical analysis methodologies.

Index Terms: Handwriting, Computational Vision, Matching.

1 Introduction

Offline writer identification plays a crucial role in various fields, such as the recognition of documents by historical authors and forensic analysis. However, its implementation presents significant challenges due to unique writing characteristics of each individual. This study aims to create a representation of images containing handwritten text by different authors and use it to make a comparison between a free text with their signatures. The signatures will include one from the same author of the free text, one from a second author, and two mutual imitations, so four in total. We will propose two methods based on visual feature matching designed to achieve effective results with a relatively small dataset. The objective is to determine which of the signatures shares the most similar pattern with the free text and, thus, if the author of the text is the same author of the said signature. We will also evaluate the accuracy by executing the algorithms on the imitated signatures.

2 Background

Deep learning models represent the state of the art in writer offline identification. While these models perform exceptionally well, they often require a large amount of data to avoid overfitting. Collecting and annotating such extensive datasets is often challenging and resource-intensive. Overfitting becomes a significant concern when the dataset is small, leading to poor generalization on unseen data, as in our case.

3 Methods

This section provides a description of the process followed for analyzing text and signatures.

3.1 Data overview

Our dataset includes 21 handwritten pages (12 from the first author and 9 from the second), taken as a reference for recognition with 4 signatures; one from the first author, one from the second author, and two signatures imitated mutually. The goal is to establish the ability of the algorithms to determine the ownership of the signatures.

3.2 Pre-processing

To enable the comparison between the text and the various signatures, we first feed the images to a filtering procedure to reduce noise and improve definition.

- **Low Pass Filtering**: A low pass filter was applied to reduce noise present in the image by allowing low frequencies to pass. In this paper, a Gaussian filter has been used.
- **High Pass Filtering**: Afterwards, a high pass filter is applied to further emphasize the characteristics of the text. This type of filter is particularly useful for highlighting finer details and improving the definition of the text.
- **Canny Filter**: In parallel with the low pass and high pass filters, the Canny filter is applied to evaluate its effectiveness in improving the definition and contrast of the text. The Canny filter is known for its ability to detect the edges of objects in the image.
- **Variable thresholding**: Another way to obtain text segmentation is thresholding. Variable thresholding suites this task particularly well.

3.3 Feature extraction

The second fundamental step was to extract the descriptors of the text and signature features so that they could be compared later. The algorithm selected for the extraction of local features is the Scale-Invariant Feature Transform (SIFT) [1].

3.4 Feature Matching

In this first method, we match the keypoints computed in all the signatures with the ones in the free text, then check which of the signature has obtained the best ratio $\frac{\texttt{\#good matches}}{\texttt{\#total matches}}$. This algorithm compares the descriptors of the signatures to each descriptor of the text. Every keypoint in the text has a correspondent first and second closest keypoint in the signature, those are the two best matches.
Then, the so called "ratio test" ([1], chapter 7.1) is applied: if the distance of the best match is less than 70% of the distance of the second-best match, the match is considered "good" and then kept, discarded otherwise.

3.5 Bag of words

In this other approach, we have used the Bag of Words representation to save text patterns and details in a codebook/dictionary. We grouped descriptors within clusters through the k-means. This allowed the representation of each image pattern in the form of a histogram/frequency vector, so similar histograms correspond to similar visual word in the codebook. Consequently, we can verify which signatures correspond to the reference text through cosine similarity applied on the frequency vectors. We decided to use cosine similarity as it focuses more on the "direction" of the vectors, instead of their length. Intuitively, a high cosine similarity implies that the two BoW vectors share the same patterns/features. For example, using Euclidean distance might not be accurate because two images with similar overall patterns but different keypoint frequencies could still have a large Euclidean distance.

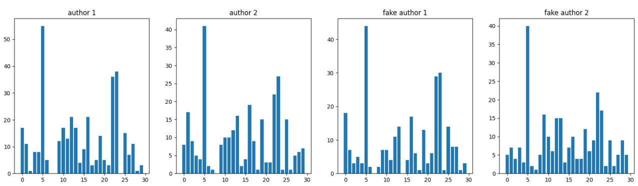

Figure 1. Visualization of the BoW clusters of the signatures

4 Experiments

In this section, we provide details on the implementation of different techniques used in our study and present the obtained results.

4.1 Experiment details

During this project, we analyzed texts produced by two different authors, along with four signatures: one for each author and two reciprocal imitations. We applied two different image filtering methods: Canny and Threshold. One of us thought that the Canny approach would highlight significant details in the image, while the other sustained that the thresholding technique would be better suited for text segmentation.

Figure 2. Same signature pre-processed with canny (top) and with variable thresholding (bottom)

Our hypothesis was partially confirmed: using the Scale-Invariant Feature Transform algorithm to calculate the descriptors (Fig. 3, Fig. 4) of the filtered images, we obtained a lower number of descriptor matches but with greater accuracy in recognizing the true authors of signatures and texts.

Figure 3. Keypoints on the signatures. In order from top left to bottom right, true signature of the first author, true signature of the second author, fake signature of the first author from the second and fake signature of the second author from the first

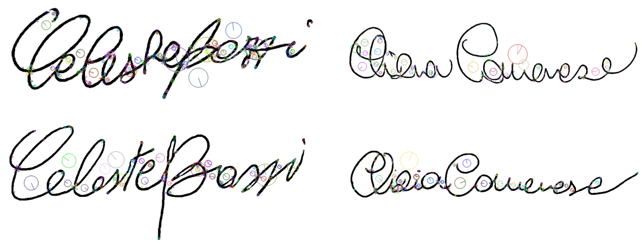

Figure 4. Same as Figure [3], but with adaptive threshold

4.1.1 Feature matching

To evaluate the effectiveness of feature matching, we introduced the following indicator:

$$match\ score = \frac{\texttt{\#good matches}}{\texttt{\#total matches}}$$

This parameter allowed us to quantify how much the Canny filter improved the distinction of the writing characteristics of the various authors (Fig. 5, Fig. 6).

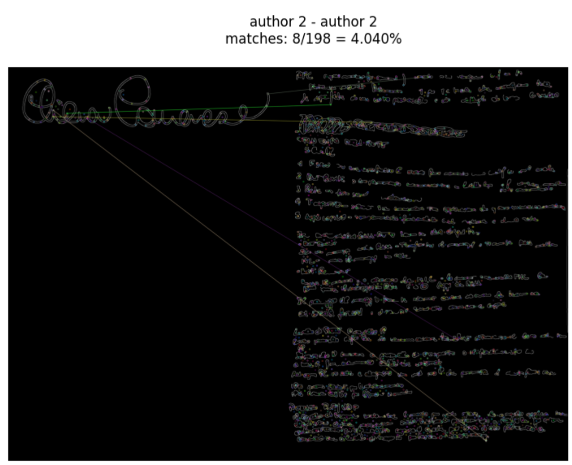

Figure 5. Best result obtained on the canny pre-processed version. The author of the signature is the same as the author of the text.

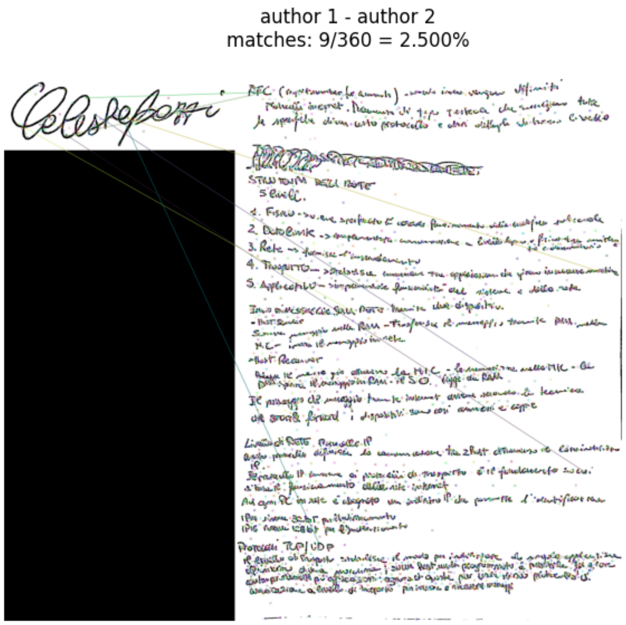

Figure 6. Best result obtained on the thresholded version. The author of the signature is NOT the same as the author of the text.

To fully evaluate the efficiency of this method, we compared each signature with each text, using the overall error as a metric to assess the matching results. To filter out lower matching values, we set a threshold of 2% for correct matches in the pre-processed version with the Canny filter and a threshold of 3% in the version with adaptive thresholding. The results of the experiment are shown in Table 1 and Table 2.

Table 1
Matching results with Canny pre-processing

Signature	Match Score
author 2 → author1_12.png	4.040%
author 1 → author1_7.png	3.252%
author 1 → author1_10.png	2.614%
author 2 → author1_6.png	2.525%
author 2 → author2_8.png	2.459%
author 2 → author1_8.png	2.459%
author 2 → author1_9.png	2.459%
author 1 → author1_6.png	2.439%
author 2 → author1_1.png	2.020%
author 2 → author1_10.png	2.020%

error on top 2%: 60%.

Table 2
Matching results with Threshold pre-processing

Signature	Match Score
author 1 → author2_6.png	5.725%
author 2 → author1_7.png	5.166%
author 2 → author2_5.png	4.428%
author 2 → author2_7.png	4.240%
author 2 → author1_10.png	3.690%
author 1 → author1_2.png	3.435%
author 2 → author2_9.png	3.321%
author 2 → author2_2.png	3.321%
author 2 → author2_5.png	3.180%
author 2 → author1_6.png	3.180%
author 2 → author1_10.png	3.180%
author 1 → author2_9.png	3.053%
author 1 → author1_6.png	3.053%

error on top 3%: 42.86%

In general, in the variable thresholding the match score is higher, and the error is lower.
The matching performed using the first method, brute force, did not produce satisfactory results, prompting us to adopt the Bag of Words approach.

4.1.2 Bag of Words

Best K selection for K-Means In the Bag of Words method, the choice of the best number of cluster centers is an important variable that influence the final results. For this reason, the best number of cluster centroids has been chosen by trying different codebooks with a varying number of cluster centroids, from 10 to 210. The best number of centroids is the one that minimizes the average error.
Different dictionaries with the given number of clusters are created, and for each of these dictionaries, the cosine similarity is calculated between the signatures frequency vector and the previously created text frequency vector. Subsequently, wrong matches are counted—i.e., if a signature has maximum similarity with another author's text—and finally, $\frac{\texttt{\#wrong matches}}{\texttt{\#total matches}}$ is calculated. In this case, there are two types of errors: the error from signature to text and the error from text to signature. The average error represents the average of the signature and text error and is calculated as $= \frac{err_{sign} \cdot 1.1 + err_{text}}{2}$. We decided to weight the signature error more due to having fewer signatures compared to texts, so misclassifying one of them should weigh more in the error calculation.

For the Bag of Words approach, we used the following parameters:

$$Precision = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

$$Recall = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

$$F1\ score = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where a true positive is "the signature/text is from author 1 and the matched text/signature is from author 1". We evaluate these metrics only onto the text-to-sign match scenario, as in the other case, due to scarcity of signatures, these metrics are not significant.

Table 3
Results with single author BoW: Canny pre-processing

Signature	Results
author 1 → author2_4.png	×
author 2 → author2_2.png	✓
fake author 1 → author2_2.png	✓
fake author 2 → author2_2.png	×

$$error_{signs} = 50\%$$

Table 4
Results with single author BoW: Texts with Canny pre-processing

Text	Results
author2_8.png → author 1	×
author2_4.png → fake author 1	✓
author2_5.png → author 1	×
author2_9.png → author 1	×
author2_6.png → fake author 1	✓
author2_1.png → fake author 1	✓
author2_2.png → fake author 1	✓
author2_7.png → fake author 1	✓
author2_3.png → fake author 1	✓

$$error_{texts} = 33.33\%$$

$$Precision = 50\%$$

$$Recall = 33.33\%$$

$$F1\ score = 40\%$$

Table 5
Results with single author BoW: Thresholding pre-processing

Signature	Results
author 1 → author2_6.png	×
author 2 → author2_6.png	✓
fake author 1 → author2_6.png	✓
fake author 2 → author2_6.png	×

$$error_{signs} = 50\%$$

Table 6
Results with single author BoW: Texts with Thresholding pre-processing

Text	Results
author2_8.png → author 2	✓
author2_4.png → author 2	✓
author2_5.png → fake author 2	×
author2_9.png → author 2	✓
author2_6.png → fake author 2	×
author2_1.png → fake author 2	×
author2_2.png → author 2	✓
author2_7.png → author 2	✓
author2_3.png → fake author 2	×

$$error_{texts} = 44.44\%$$

$$Precision = 50\%$$

$$Recall = 27.78\%$$

$$F1\ score = 35.71\%$$

Results of the Bag of Words From the results reported in tables 3, 4, 5 and 6, we observe that the Bag of Words algorithm, where the dictionary was constructed using texts from only one author, author 2, performs better when the Canny filter is used in pre-processing. This is because the Canny filter captures structural details of the image more effectively, thus our dictionary representation includes, although limited, well-defined characteristics of a specific author's handwriting. This results in better performance compared to using thresholding.
On the other hand, as we can see from the values in tables 7, 8, 9 and 10, when more data is present, the dictionary is enriched with features from both authors, covering a broader spectrum of possible characteristics. This diversity compensates for the loss of fine details due to thresholding. Therefore, in this case, thresholding yields better results as a broader vocabulary can interpret and match simplified patterns in the images more effectively.

Table 7
Results with multi author BoW: Canny pre-processing

Signature	Results
author 1 → author1_12.png	✓
author 2 → author1_1.png	×
fake author 1 → author2_2.png	✓
fake author 2 → author1_1.png	✓

$$error_{signs} = 25\%$$

Table 8
Results with multi author BoW: Texts with Canny pre-processing

Text	Results
author2_8.png → author 1	×
author1_1.png → author 1	✓
author2_4.png → author 1	×
author2_5.png → author 1	×
author1_11.png → author 1	✓
author2_9.png → author 1	×
author1_2.png → author 1	✓
author1_4.png → author 1	✓
author2_6.png → author 1	×
author1_12.png → author 1	✓
author1_7.png → author 1	✓
author1_8.png → author 1	✓
author2_1.png → author 1	×
author2_2.png → author 1	×
author2_7.png → author 1	×
author1_3.png → author 1	✓
author1_5.png → author 1	✓
author1_6.png → author 1	✓
author1_10.png → author 1	✓
author2_3.png → author 1	×
author1_9.png → author 1	✓

$$error_{texts} = 42.86\%$$

$$Precision = 28.57\%$$

$$Recall = 50\%$$

$$F1\ score = 36.36\%$$

Table 9
Results with multi author BoW: Thresholding pre-processing

Signature	Results
author 1 → author2_6.png	×
author 2 → author2_6.png	✓
fake author 1 → author2_6.png	✓
fake author 2 → author2_6.png	×

$$error_{signs} = 50\%$$

Table 10
Results with multi author BoW: Texts with Thresholding pre-processing

Text	Results
author2_8.png → author 2	✓
author1_1.png → fake author 2	✓
author2_4.png → author 2	✓
author2_5.png → author 2	✓
author1_11.png → fake author 2	✓
author2_9.png → author 2	✓
author1_2.png → fake author 2	✓
author1_4.png → fake author 2	✓
author2_6.png → author 2	✓
author1_12.png → fake author 2	✓
author1_7.png → fake author 2	✓
author1_8.png → fake author 2	✓
author2_1.png → author 2	✓
author2_2.png → author 2	✓
author2_7.png → author 2	✓
author1_3.png → fake author 2	✓
author1_5.png → fake author 2	✓
author1_6.png → fake author 2	✓
author1_10.png → fake author 2	✓
author2_3.png → author 2	✓
author1_9.png → fake author 2	✓

$$error_{texts} = 0\%$$

$$Precision = 100\%$$

$$Recall = 100\%$$

$$F1\ score = 100\%$$

5 Conclusion

This study brought up two different computer vision based methods to tackle the problem of writer identification with a small dataset, applied on different image representations and pre-processings. If the texts dataset mainly contains samples written by the same author, Canny pre-processing and Bag of Words approach is the one that performs best, but in absolute units the results are just a little better than chance. If the texts dataset is richer than only one author, variable thresholding representation and Bag of Words is the right choice, and in proportion much better than Canny - BoW in the single writer dictionary scenario. We also enlightened that the signature error in BoW is always greater than the texts error, that is due to the fact that the dataset contains only 4 signatures and 21 texts.

6 Future works

Alternative possibilities to improve and deepen this work could include the analysis of the algorithms used but with a greater number of signatures, in order to be able to verify the change in the error with an increase in data. Similarly, it would be interesting to extend these methods to include more than two authors. Exploring these directions could lead to significant developments in signature verification, allowing us to better understand the capabilities and limitations of existing algorithms in more complex and realistic scenarios.

References

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